

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2022 P2HR Worked Solutions

Jun 2022 | P2HR

31 questions ■ question image followed by worked solution

PREPARED BY

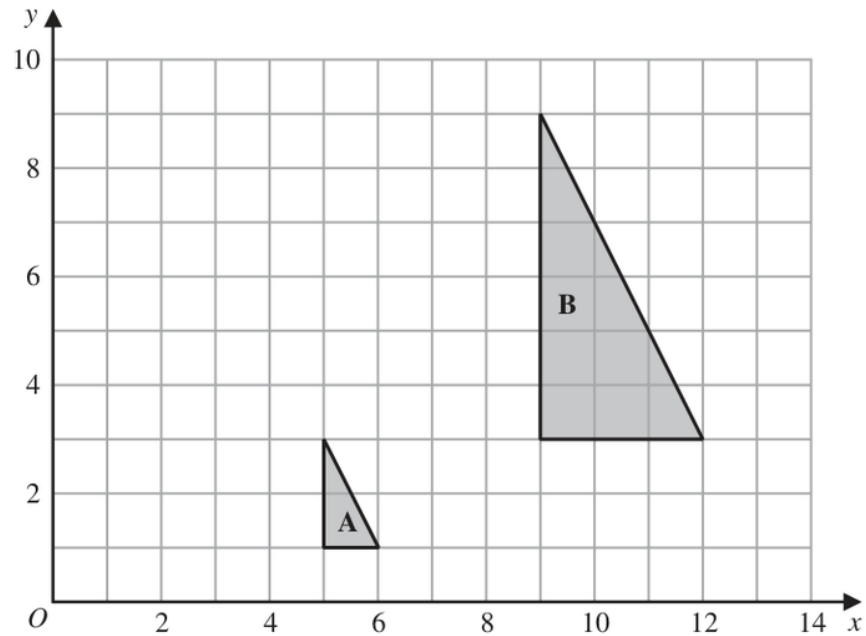
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PAST PAPER QUESTION**Q#: 1****Marks: 4****TOPIC: Transformations**

Jun 2022 P2HR - Jun 2022 | P2HR

1

(a) Describe fully the single transformation that maps triangle **A** onto triangle **B**

(3)

(b) On the grid above, translate triangle **A** by the vector $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$

Label your triangle **C**

(1)

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION**Q#: 1****Marks: 4****TOPIC: Transformations**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**(a) Triangle A is enlarged to triangle B .**2. The scale factor is**

The scale factor is

$$3$$

3. Use Transformations tag correctUsing corresponding vertices gives centre $(3, 0)$.**4. Use Transformations tag correct**So the transformation is an enlargement, scale factor 3, centre $(3, 0)$.**5. (b) Translating triangle by**(b) Translating triangle A by

$$\begin{pmatrix} -4 \\ 3 \end{pmatrix}$$

6. gives vertices

gives vertices

$$(1, 4), (2, 4), (1, 6)$$

FINAL ANSWER(a) enlargement, scale factor 3, centre $(3, 0)$; (b) vertices $(1, 4), (2, 4), (1, 6)$.

PAST PAPER QUESTION**Q#: 2****Marks: 3****TOPIC: Prime Factors, HCF & LCM**

Jun 2022 P2HR - Jun 2022 | P2HR

- 2** Write 1200 as a product of powers of its prime factors.
Show your working clearly.

(Total for Question 2 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 2

Marks: 3

TOPIC: Prime Factors, HCF & LCM

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Find the LCM

$$1200 = 12 \times 100$$

$$1200 = (2^2 \times 3)(2^2 \times 5^2)$$

$$1200 = 2^4 \times 3 \times 5^2$$

FINAL ANSWER

$$2^4 \times 3 \times 5^2$$

EA

PAST PAPER QUESTION**Q#: 3****Marks: 3****TOPIC: Statistics Toolkit**

Jun 2022 P2HR - Jun 2022 | P2HR

3 Alberto, Bill, Candela and Diana are four friends.

Here is some information about the height of each of these friends.

Alberto's height is 158 cm.

Bill's height is 175 cm.

Candela's height is greater than Diana's height.

The median height of these four friends is 160 cm.

The range of the heights of these four friends is 21 cm.

Work out Candela's height and Diana's height.

Candela cm

Diana cm

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 3**TOPIC: **Statistics Toolkit**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The median is . Since**

The median is 160. Since the ordered heights are

Diana, 158, Candela, 175

$$\frac{158 + \text{Candela}}{2} = 160$$

$$\text{Candela} = 162$$

2. The range is

The range is 21:

$$175 - \text{Diana} = 21$$

$$\text{Diana} = 154$$

FINAL ANSWER

Candela = 162 cm, Diana = 154 cm.

PAST PAPER QUESTION**Q#: 4****Marks: 5****TOPIC: Set Notation & Venn Diagrams**

Jun 2022 P2HR - Jun 2022 | P2HR

4 $\mathcal{E} = \{9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$ $A = \{\text{multiples of 3}\}$ $B = \{\text{odd numbers}\}$

(a) List the members of the set

(i) $A \cap B$
(1)(ii) $A \cup B$
(1)(b) Is it true that $24 \in A$?

Tick one of the boxes below.

Yes

No

☐☐

Give a reason for your answer.

.....
(1)Set C has 4 members such that $C \cap B' = \{10, 18\}$ (c) List the members of one possible set C
(2)**(Total for Question 4 is 5 marks)**

PAST PAPER SOLUTION**Q#: 4****Marks: 5****TOPIC: Set Notation & Venn Diagrams**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$\mathcal{E} = \{9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$$

2. The multiples of in areThe multiples of 3 in $\mathcal{E}$ are

$$A = \{9, 12, 15, 18\}$$

3. The odd numbers in areThe odd numbers in $\mathcal{E}$ are

$$B = \{9, 11, 13, 15, 17, 19\}$$

4. Calculate value

So

$$A \cap B = \{9, 15\}$$

5. Calculate value

and

$$A \cup B = \{9, 11, 12, 13, 15, 17, 18, 19\}$$

6. The statement is false becauseThe statement $24 \in A$ is false because 24 is not in the universal set $\mathcal{E}$.**7. Calculate value**For C , we need $C \cap B' = \{10, 18\}$ and C has 4 members. One valid choice is

$$C = \{9, 10, 11, 18\}$$

FINAL ANSWER(a)(i) $\{9, 15\}$, (a)(ii) $\{9, 11, 12, 13, 15, 17, 18, 19\}$, (b) No, (c) one possible set is $\{9, 10, 11, 18\}$

PAST PAPER QUESTION**Q#: 5****Marks: 4****TOPIC: Area & Perimeter**

Jun 2022 P2HR - Jun 2022 | P2HR

- 5 The diagram shows a shape made from a square $ABCD$ and 4 identical semicircles.

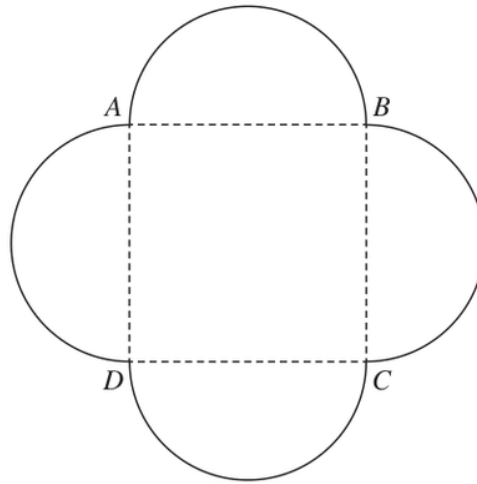


Diagram **NOT**
accurately drawn

As shown in the diagram, the semicircles have AB , BC , CD and DA as diameters.

The area of the square is 36 cm^2

Calculate the total area of the shape.

Give your answer correct to one decimal place.

..... cm^2

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 4****TOPIC: Area & Perimeter**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate area**

The square has area 36 cm^2 , so its side length is

$$\sqrt{36} = 6 \text{ cm}$$

2. Calculate area

Each semicircle has diameter 6 cm, so radius

$$r = 3 \text{ cm}$$

3. Area of 4 semicircles

Area of 4 semicircles:

$$4 \times \frac{1}{2} \pi (3)^2 = 18\pi$$

4. Total area

Total area:

$$36 + 18\pi = 92.548 \dots$$

FINAL ANSWER

92.5 cm^2 .

PAST PAPER QUESTION**Q#: 6****Marks: 8****TOPIC: Solving Linear Equations**

Jun 2022 P2HR - Jun 2022 | P2HR

6 (a) Solve $p = \frac{3p - 5}{10}$

Show clear algebraic working.

$$p = \dots\dots\dots (3)$$

(b) Simplify a^0 where $a > 0$

$$\dots\dots\dots (1)$$

(c) Simplify fully $\frac{3xy^3}{6x^2y}$

$$\dots\dots\dots (2)$$

(d) Factorise fully $10c^3d^2 + 15cd^4$

$$\dots\dots\dots (2)$$

(Total for Question 6 is 8 marks)

PAST PAPER SOLUTION**Q#: 6** **Marks: 8****TOPIC: Solving Linear Equations**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$p = \frac{3p - 5}{10}$$

$$10p = 3p - 5$$

$$7p = -5$$

$$p = -\frac{5}{7}$$

2. (b)(b) $a^0 = 1$.**3. (c)**

(c)

$$\frac{3xy^3}{6x^2y} = \frac{y^2}{2x}$$

4. (d)

(d)

$$10c^3d^2 + 15cd^4 = 5cd^2(2c^2 + 3d^2)$$

FINAL ANSWER(a) $p = -\frac{5}{7}$, (b) 1, (c) $\frac{y^2}{2x}$, (d) $5cd^2(2c^2 + 3d^2)$

PAST PAPER QUESTION**Q#: 7****Marks: 2****TOPIC: Algebraic Roots & Indices**

Jun 2022 P2HR - Jun 2022 | P2HR

7 $\frac{2^k}{4^n} = 2^x$

Find an expression for x in terms of k and n

$x = \dots\dots\dots$

(Total for Question 7 is 2 marks)

EA

PAST PAPER SOLUTION**Q#: 7****Marks: 2****TOPIC: Algebraic Roots & Indices**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use index laws**

$$\frac{2^k}{4^n} = 2^x$$

2. Use index lawsSince $4 = 2^2$,

$$4^n = (2^2)^n = 2^{2n}$$

3. Use index laws

So

$$\frac{2^k}{2^{2n}} = 2^{k-2n}$$

4. Use index laws

Therefore

$$x = k - 2n$$

FINAL ANSWER

$$x = k - 2n.$$

PAST PAPER QUESTION**Q#: 8****Marks: 3**TOPIC: **Percentages**

Jun 2022 P2HR - Jun 2022 | P2HR

- 8** A cinema increased the cost of an adult ticket by 12%
- After the increase, the cost of an adult ticket was £18.20
- Work out the cost of an adult ticket before the increase.

£.....

(Total for Question 8 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 8****Marks: 3****TOPIC: Percentages**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate percentage**

After a 12% increase, the ticket price is 112% of the original price.

$$18.20 = 1.12 \times \text{original price}$$

$$\text{original price} = \frac{18.20}{1.12} = 16.25$$

FINAL ANSWER**£16.25**

EA

PAST PAPER QUESTION**Q#: 9****Marks: 4****TOPIC: Powers, Roots & Standard Form**

Jun 2022 P2HR - Jun 2022 | P2HR

- 9 The table gives information about the population, correct to 2 significant figures, of each of five cities in 2018

City	Population (2018)
Ahmedabad	7.7×10^6
Barcelona	5.5×10^6
Chicago	8.8×10^6
Lagos	1.3×10^7
Tokyo	3.7×10^7

- (a) Write 8.8×10^6 as an ordinary number.

.....
(1)

- (b) Which of these cities had the least population in 2018?

.....
(1)

- (c) Work out the difference between the population of Tokyo and the population of Ahmedabad in 2018

Give your answer in standard form correct to 2 significant figures.

.....
(2)

(Total for Question 9 is 4 marks)

PAST PAPER SOLUTION**Q#: 9** **Marks: 4****TOPIC: Powers, Roots & Standard Form**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Convert standard form**

$$8.8 \times 10^6 = 8800000$$

2. The least population in the

The least population in the table is Barcelona, with

$$5.5 \times 10^6$$

3. Convert standard form

For the difference between Tokyo and Ahmedabad:

$$3.7 \times 10^7 - 7.7 \times 10^6$$

$$37000000 - 7700000 = 29300000$$

$$29300000 = 2.93 \times 10^7$$

4. Correct to 2 significant figures

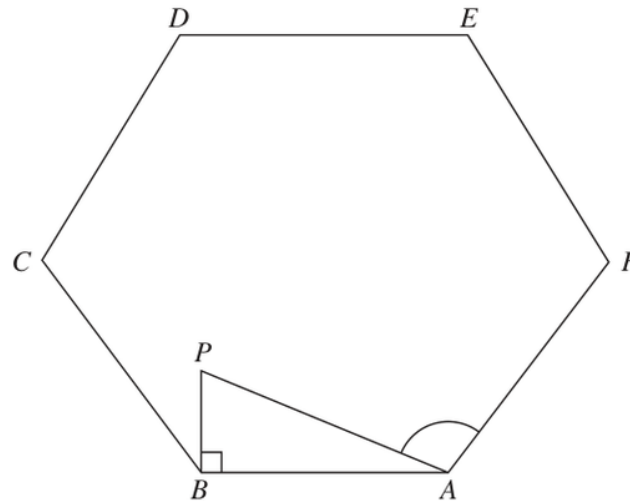
Correct to 2 significant figures:

$$2.9 \times 10^7$$

FINAL ANSWER(a) 8800000, (b) Barcelona, (c) 2.9×10^7

PAST PAPER QUESTION**Q#: 10****Marks: 5****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2022 P2HR - Jun 2022 | P2HR

10 The diagram shows triangle ABP inside the regular hexagon $ABCDEF$ Diagram **NOT**
accurately drawn

$AB = 5 \text{ cm}$

$BP = 2 \text{ cm}$

$\text{Angle } ABP = 90^\circ$

Work out the size of angle PAF

Give your answer correct to 3 significant figures.

o

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION**Q#: 10****Marks: 5****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. A regular hexagon has interior**

A regular hexagon has interior angle

$$120^\circ$$

2. Use trigonometryIn right-angled triangle ABP ,

$$\tan(\angle PAB) = \frac{BP}{AB} = \frac{2}{5}$$

$$\angle PAB = \tan^{-1}\left(\frac{2}{5}\right) = 21.801\dots^\circ$$

3. Use trigonometry

Therefore

$$\angle PAF = 120^\circ - 21.801\dots^\circ = 98.198\dots^\circ$$

FINAL ANSWER

98.2°.

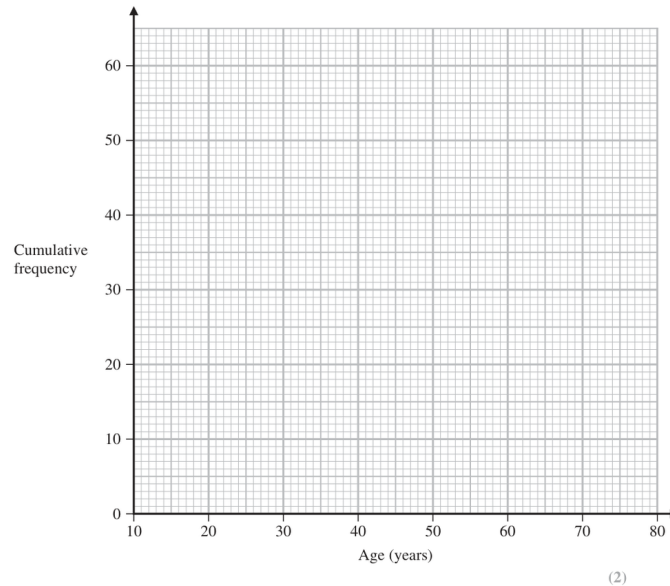
PAST PAPER QUESTION**Q#: 11****Marks: 7****TOPIC: Cumulative Frequency Diagrams**

Jun 2022 P2HR - Jun 2022 | P2HR

- 11 The cumulative frequency table shows information about the ages of 60 people who went to a gym on Saturday.

Age (a years)	Cumulative frequency
$10 < a \leq 20$	13
$10 < a \leq 30$	36
$10 < a \leq 40$	42
$10 < a \leq 50$	47
$10 < a \leq 60$	52
$10 < a \leq 70$	56
$10 < a \leq 80$	60

- (a) On the grid, draw a cumulative frequency graph for the information in the table.

**Question 11 continued**

- (b) Use your graph to find an estimate for the median of the ages of these people.

..... years
(1)

- (c) Use your graph to find an estimate for the interquartile range of the ages of these people.

..... years
(2)

- (d) Use your graph to find an estimate for the number of these people who are older than 55 years.

.....
(2)

(Total for Question 11 is 7 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 7****TOPIC: Cumulative Frequency Diagrams**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Plot the points**

Plot the points

$$(10, 0), (20, 13), (30, 36), (40, 42), (50, 47), (60, 52), (70, 56), (80, 60)$$

2. The median is at cumulative

The median is at cumulative frequency 30.

3. Convert standard form

This is between (20, 13) and (30, 36):

$$20 + \frac{30 - 13}{36 - 13} \times 10 = 27.39 \dots$$

4. Median age is about years

Median age is about 27.4 years.

5. Convert standard form

For the interquartile range:

$$Q_1 \text{ is at } 15, \quad Q_3 \text{ is at } 45$$

$$Q_1 \approx 20 + \frac{15 - 13}{36 - 13} \times 10 = 20.9$$

$$Q_3 \approx 40 + \frac{45 - 42}{47 - 42} \times 10 = 46$$

$$\text{IQR} \approx 46 - 20.9 = 25.1$$

6. Use cumulative frequency

For people older than 55:

$$F(55) \approx 47 + \frac{5}{10}(52 - 47) = 49.5$$

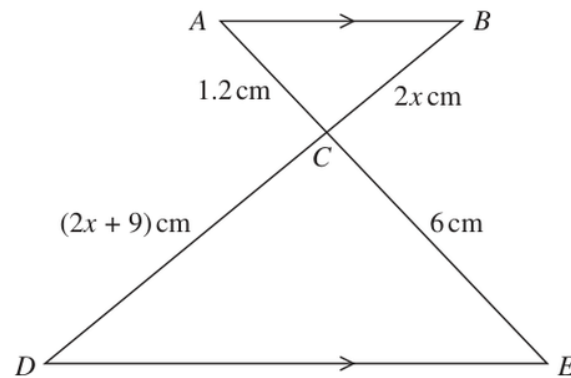
$$60 - 49.5 = 10.5$$

FINAL ANSWER

median about 27.4 years, IQR about 25.1 years, about 11 people older than 55.

PAST PAPER QUESTION**Q#: 12****Marks: 3****TOPIC:** Congruence, Similarity & Geometrical Proof

Jun 2022 P2HR - Jun 2022 | P2HR

12Diagram **NOT**
accurately drawn

ACE and BCD are straight lines.
 AB is parallel to DE

Work out the value of x

$x = \dots\dots\dots$

(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION**Q#: 12****Marks: 3****TOPIC:** Congruence, Similarity & Geometrical Proof

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the gradient**

Since AB is parallel to DE , triangles ACB and ECD are similar.

2. Split the ratio

So corresponding sides are in the same ratio:

$$\frac{AC}{CE} = \frac{BC}{CD}$$

3. Substitute the lengths

Substitute the lengths:

$$\frac{1.2}{6} = \frac{2x}{2x + 9}$$

$$\frac{1}{5} = \frac{2x}{2x + 9}$$

$$2x + 9 = 10x$$

$$8x = 9$$

$$x = \frac{9}{8}$$

FINAL ANSWER

$$x = \frac{9}{8}.$$

PAST PAPER QUESTION**Q#: 13****Marks: 3****TOPIC: Circles, Arcs & Sectors**

Jun 2022 P2HR - Jun 2022 | P2HR

13 The diagram shows a sector AOB of a circle with centre O

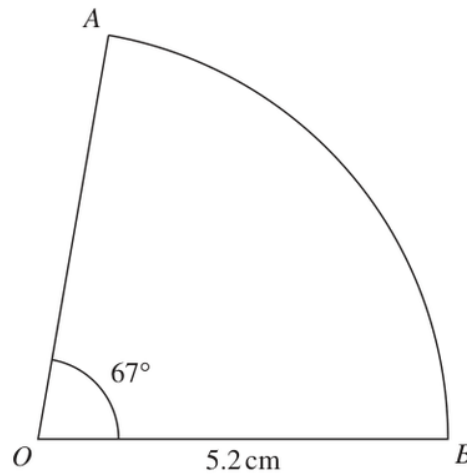


Diagram **NOT**
accurately drawn

Angle $AOB = 67^\circ$
 $OA = OB = 5.2 \text{ cm}$

Calculate the perimeter of the sector.
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13

Marks: 3

TOPIC: **Circles, Arcs & Sectors**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Arc length

Arc length:

$$\frac{67}{360} \times 2\pi(5.2) = 6.080 \dots$$

2. Perimeter

Perimeter:

$$5.2 + 5.2 + 6.080 \dots = 16.480 \dots$$

FINAL ANSWER

16.5 cm.

PAST PAPER QUESTION**Q#: 14****Marks: 3****TOPIC: Combined & Conditional Probability**

Jun 2022 P2HR - Jun 2022 | P2HR

14 Ciara throws **four** fair six-sided dice.

The faces of each dice are labelled with the numbers 1, 2, 3, 4, 5, 6

Work out the probability that at least one of the dice lands on an even number.

(Total for Question 14 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 14

Marks: 3

TOPIC: **Combined & Conditional Probability**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. The complement of at least

The complement of "at least one even number" is "both numbers are odd".

$$P(\text{odd}) = \frac{1}{2}$$

$$P(\text{both odd}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

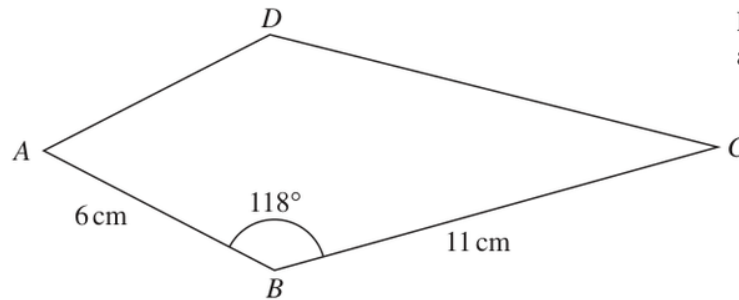
$$P(\text{at least one even}) = 1 - \frac{1}{4} = \frac{3}{4}$$

FINAL ANSWER

$$\frac{3}{4}.$$

PAST PAPER QUESTION**Q#: 15****Marks: 3****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jun 2022 P2HR - Jun 2022 | P2HR

15 The diagram shows a kite $ABCD$ 

$AB = 6 \text{ cm}$

$BC = 11 \text{ cm}$

$\text{Angle } ABC = 118^\circ$

Calculate the area of the kite.

Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION**Q#: 15****Marks: 3****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. A kite can be split**

A kite can be split into two congruent triangles.

2. Area of triangleArea of triangle ABC :

$$\frac{1}{2} \times 6 \times 11 \times \sin 118^\circ$$

3. Use trigonometry

So the area of the kite is

$$\begin{aligned} 2 \times \frac{1}{2} \times 6 \times 11 \times \sin 118^\circ \\ = 6 \times 11 \times \sin 118^\circ = 58.274 \dots \end{aligned}$$

FINAL ANSWER58.3 cm².

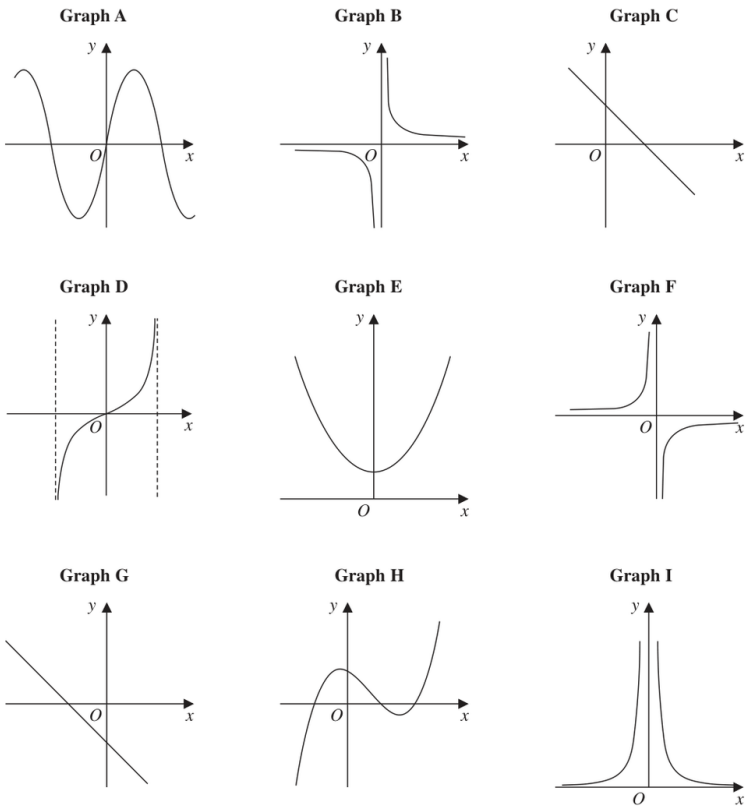
PAST PAPER QUESTION

Q#: 16 Marks: 3

TOPIC: **Graphs of Functions**

Jun 2022 P2HR - Jun 2022 | P2HR

16 Here are nine graphs.



Complete the table below with the letter of the graph that could represent each given equation.
Write each answer on the dotted line.

Equation	Graph
$y = -2x + 3$	
$y = -\frac{1}{x}$	
$y = \tan x^\circ$	
$y = (x + 1)(x - 1)(x - 2)$	

(Total for Question 16 is 3 marks)

PAST PAPER SOLUTION**Q#: 16****Marks: 3****TOPIC: Graphs of Functions**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. is a negative straight line**

$$y = -2x + 3$$

is a negative straight line with positive y -intercept, so it is **Graph C**.

2. Read the graph

$$y = \frac{1}{x}$$

is the reciprocal graph in quadrants I and III, so it is **Graph B**.

3. has repeating branches with vertical

$$y = \tan x$$

has repeating branches with vertical asymptotes and passes through the origin, so it is **Graph D**.

4. Read the graph

$$y = (x + 1)(x - 1)(x - 2)$$

is a positive cubic with three roots, so it is **Graph H**.

FINAL ANSWER

C, B, D, H.

PAST PAPER QUESTION**Q#: 17****Marks: 2****TOPIC: Algebraic Proof**

Jun 2022 P2HR - Jun 2022 | P2HR

17 Use algebra to show that $0.3\dot{4}\dot{5} = \frac{19}{55}$

(Total for Question 17 is 2 marks)

EA

PAST PAPER SOLUTION**Q#: 17****Marks: 2****TOPIC: Algebraic Proof**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let**

Let

$$x = 0.3\dot{4}\dot{5}$$

2. Calculate value

Then

$$10x = 3.4\dot{5}$$

3. Calculate value

and

$$1000x = 345.4\dot{5}$$

4. Subtract

Subtract:

$$1000x - 10x = 345.4\dot{5} - 3.4\dot{5}$$

$$990x = 342$$

$$x = \frac{342}{990} = \frac{19}{55}$$

FINAL ANSWER

$$0.3\dot{4}\dot{5} = \frac{19}{55}$$

PAST PAPER QUESTION**Q#: 18****Marks: 4****TOPIC: Rounding, Estimation & Bounds**

Jun 2022 P2HR - Jun 2022 | P2HR

18 Kaidan and Sonja went on two different car journeys.

For Kaidan's journey

distance = 80 km correct to the nearest 5 km

time = 2.7 hours correct to 1 decimal place

For Sonja's journey

distance = 33 km correct to 2 significant figures

time = 1 hour correct to the nearest 0.1 hour

Kaidan says,

"My average speed could have been greater than Sonja's average speed."

By considering bounds, show that Kaidan is correct.

Show your working clearly.

(Total for Question 18 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 18****Marks: 4****TOPIC: Rounding, Estimation & Bounds**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. To show Kaidan's average**

To show Kaidan's average speed could be greater, compare Kaidan's greatest possible speed with Sonja's smallest possible speed.

2. Estimate the value

For Kaidan:

$$d < 82.5, \quad t \geq 2.65$$

$$\text{greatest speed} = \frac{82.5}{2.65} = 31.13 \dots \text{ km/h}$$

3. Estimate the value

For Sonja:

$$d \geq 32.5, \quad t < 1.05$$

$$\text{smallest speed} = \frac{32.5}{1.05} = 30.95 \dots \text{ km/h}$$

4. Estimate the value

Since

$$31.13 \dots > 30.95 \dots$$

5. Kaidan's average speed could

Kaidan's average speed could have been greater.

FINAL ANSWER

Kaidan is correct.

PAST PAPER QUESTION**Q#: 19****Marks: 4****TOPIC: Functions**

Jun 2022 P2HR - Jun 2022 | P2HR

19 $f(x) = x^2 - 4$

$g(x) = 2x + 1$

Solve $fg(x) > 0$

Show clear algebraic working.

(Total for Question 19 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 19****Marks: 4****TOPIC: Functions**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve quadratic equation**

$$f(x) = x^2 - 4, \quad g(x) = 2x + 1$$

$$fg(x) = f(g(x)) = (2x + 1)^2 - 4$$

$$= 4x^2 + 4x + 1 - 4 = 4x^2 + 4x - 3$$

2. Solve

Solve

$$4x^2 + 4x - 3 > 0$$

3. Factorise

Factorise:

$$(2x + 3)(2x - 1) > 0$$

4. The critical values are

The critical values are

$$x = -\frac{3}{2}, \quad x = \frac{1}{2}$$

5. The product is positive outside

The product is positive outside the roots.

FINAL ANSWER

$$x < -\frac{3}{2} \text{ or } x > \frac{1}{2}.$$

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Coordinate Geometry**

Jun 2022 P2HR - Jun 2022 | P2HR

20 The centre O of a circle has coordinates $(4, 7)$

The point A , on the circle, has coordinates $(6, 11)$ and AOP is a diameter of the circle.

Find an equation of the tangent to the circle at the point P

(Total for Question 20 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 20

Marks: 4

TOPIC: **Coordinate Geometry**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. The centre of the circle

The centre of the circle is

$$O = (4, 7)$$

2. The point isThe point A is

$$A = (6, 11)$$

3. Find the midpointSince AOP is a diameter, O is the midpoint of AP .**4. Let**Let $P = (x, y)$.

$$\left(\frac{6+x}{2}, \frac{11+y}{2} \right) = (4, 7)$$

$$6+x=8, \quad 11+y=14$$

$$P = (2, 3)$$

5. The gradient of isThe gradient of OP is

$$\frac{3-7}{2-4} = 2$$

6. Use circle theorem

The tangent is perpendicular to the radius, so its gradient is

$$-\frac{1}{2}$$

7. Evaluate fractionUsing point $P = (2, 3)$,

$$y-3 = -\frac{1}{2}(x-2)$$

$$y = -\frac{1}{2}x + 4$$

FINAL ANSWER

$$y = -\frac{1}{2}x + 4$$

EA

PAST PAPER QUESTION

Q#: 20

Marks: 4

TOPIC: **Coordinate Geometry**

Jun 2022 P2HR - Jun 2022 | P2HR

20 The centre O of a circle has coordinates $(4, 7)$

The point A , on the circle, has coordinates $(6, 11)$ and AOP is a diameter of the circle.

Find an equation of the tangent to the circle at the point P

(Total for Question 20 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Coordinate Geometry**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The centre of the circle**

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FINAL ANSWER

$$y = -\frac{1}{2}x + 4$$

EA

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Simultaneous Equations**

Jun 2022 P2HR - Jun 2022 | P2HR

21 Solve the simultaneous equations

$$\begin{aligned}x - 2y &= 3 \\ x^2 - y^2 + 2x &= 10\end{aligned}$$

Show clear algebraic working.

(Total for Question 21 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 21****Marks: 5****TOPIC: Simultaneous Equations**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$x - 2y = 3$$

2. Calculate value

So

$$x = 2y + 3$$

3. Calculate value

Substitute into

$$x^2 - y^2 + 2x = 10$$

$$(2y + 3)^2 - y^2 + 2(2y + 3) = 10$$

$$4y^2 + 12y + 9 - y^2 + 4y + 6 = 10$$

$$3y^2 + 16y + 5 = 0$$

$$(3y + 1)(y + 5) = 0$$

4. Evaluate fraction

So

$$y = -\frac{1}{3} \quad \text{or} \quad y = -5$$

5. FindFind x :

$$y = -\frac{1}{3} \Rightarrow x = \frac{7}{3}$$

$$y = -5 \Rightarrow x = -7$$

FINAL ANSWER

$$(x, y) = \left(\frac{7}{3}, -\frac{1}{3}\right) \text{ or } (-7, -5)$$

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Simultaneous Equations**

Jun 2022 P2HR - Jun 2022 | P2HR

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$$\begin{aligned}x - 2y &= 3 \\ x^2 - y^2 + 2x &= 10\end{aligned}$$

Show clear algebraic working.

(Total for Question 21 is 5 marks)

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Jun 2022 P2HR - Jun 2022 | P2HR

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Substitute into

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FINAL ANSWER

$$(x, y) = \left(\frac{7}{3}, -\frac{1}{3}\right) \text{ or } (-7, -5)$$

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: Transformations of Graphs**

Jun 2022 P2HR - Jun 2022 | P2HR

22 The point A with coordinates $(-3, 2)$ lies on the straight line with equation $y = f(x)$

(a) Find the coordinates of the image of the point A on the straight line with equation

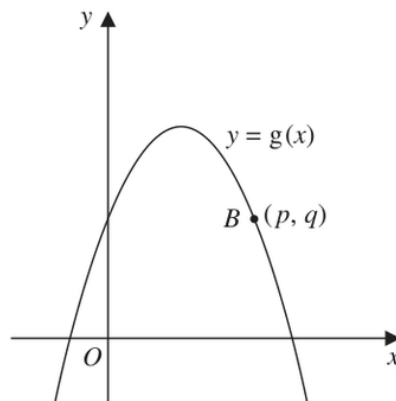
(i) $y = f(x) - 3$

(.....,)
(1)

(ii) $y = f\left(\frac{x}{2}\right)$

(.....,)
(1)

Here is a sketch of part of the curve with equation $y = g(x)$



The point B with coordinates (p, q) lies on the curve.

(b) Find the coordinates of the image of the point B on the curve with equation

$$y = -g(x - c)$$

where c is a constant.

(.....,)
(2)

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 4****TOPIC: Transformations of Graphs**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Point lies on**Point $A = (-3, 2)$ lies on $y = f(x)$.**2. Calculate value**

For

$$y = f(x) - 3$$

3. the graph moves 3 units

the graph moves 3 units down:

$$(-3, 2) \rightarrow (-3, -1)$$

4. Evaluate fraction

For

$$y = f\left(\frac{x}{2}\right)$$

5. the coordinate is multiplied bythe x -coordinate is multiplied by 2:

$$(-3, 2) \rightarrow (-6, 2)$$

6. Calculate valueFor part (b), point $B = (p, q)$ lies on $y = g(x)$.**7. Calculate value**

For

$$y = -g(x - c)$$

8. Read the graph $x - c$ moves the graph c units right, and the minus sign reflects the graph in the x -axis.**9. Apply transformation**

So

$$(p, q) \rightarrow (p + c, -q)$$

FINAL ANSWER $(-3, -1)$, $(-6, 2)$, and $(p + c, -q)$

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: Transformations of Graphs**

Jun 2022 P2HR - Jun 2022 | P2HR

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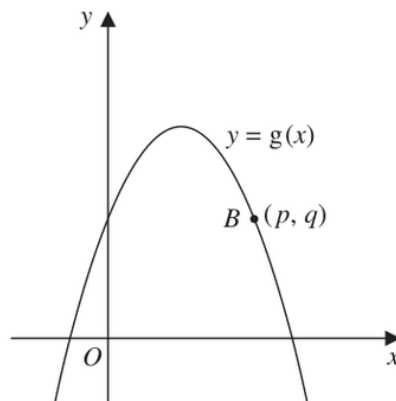
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$$y = -g(x - c)$$

where c is a constant.

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(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 4****TOPIC: Transformations of Graphs**

Jun 2022 P2HR - Jun 2022 | P2HR

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So

$$(p, q) \rightarrow (p + c, -q)$$

FINAL ANSWER $(-3, -1)$, $(-6, 2)$, and $(p + c, -q)$

PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Algebraic Fractions**

Jun 2022 P2HR - Jun 2022 | P2HR

23 Express $\left(\frac{20}{x^2 - 36} - \frac{2}{x - 6}\right) \times \frac{1}{4 - x}$ as a single fraction in its simplest form.

(Total for Question 23 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 23****Marks: 3****TOPIC: Algebraic Fractions**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**

$$\left(\frac{20}{x^2 - 36} - \frac{2}{x - 6} \right) \times \frac{1}{4 - x}$$

2. Factor

Factor:

$$x^2 - 36 = (x - 6)(x + 6)$$

3. Use trigonometry

So

$$\frac{20}{(x - 6)(x + 6)} - \frac{2}{x - 6}$$

4. Use the common denominatorUse the common denominator $(x - 6)(x + 6)$:

$$\begin{aligned} & \frac{20 - 2(x + 6)}{(x - 6)(x + 6)} \\ &= \frac{20 - 2x - 12}{(x - 6)(x + 6)} \\ &= \frac{8 - 2x}{(x - 6)(x + 6)} \\ &= \frac{-2(x - 4)}{(x - 6)(x + 6)} \end{aligned}$$

5. Use trigonometry

Now multiply by

$$\begin{aligned} & \frac{1}{4 - x} = -\frac{1}{x - 4} \\ & \frac{-2(x - 4)}{(x - 6)(x + 6)} \times \left(-\frac{1}{x - 4} \right) \end{aligned}$$

6. CancelCancel $x - 4$:

$$\frac{2}{(x - 6)(x + 6)}$$

$$= \frac{2}{x^2 - 36}$$

FINAL ANSWER

$$\frac{2}{x^2 - 36}$$

EA

PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Algebraic Fractions**

Jun 2022 P2HR - Jun 2022 | P2HR

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EA

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Jun 2022 P2HR - Jun 2022 | P2HR

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Factor:

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FINAL ANSWER

$$\frac{2}{x^2 - 36}$$

EA

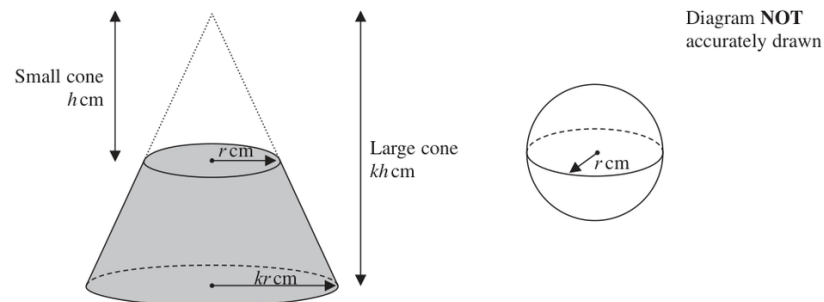
PAST PAPER QUESTION**Q#: 24****Marks: 6****TOPIC: Area & Volume of Similar Shapes**

Jun 2022 P2HR - Jun 2022 | P2HR

24 The diagram shows a frustum of a cone, and a sphere.

The frustum, shown shaded in the diagram, is made by removing the small cone from the large cone.

The small cone and the large cone are similar.



The height of the small cone is h cm and the radius of the base of the small cone is r cm.
The height of the large cone is kh cm and the radius of the base of the large cone is kr cm.
The radius of the sphere is r cm.

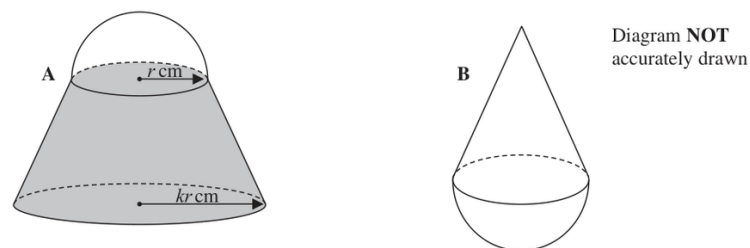
The sphere is divided into two hemispheres, each of radius r cm.

Solid A is formed by joining one of the hemispheres to the frustum.

The plane face of the hemisphere coincides with the upper plane face of the frustum, as shown in the diagram below.

Solid B is formed by joining the other hemisphere to the small cone that was removed from the large cone.

The plane face of the hemisphere coincides with the plane face of the base of the small cone, as shown in the diagram below.



The volume of solid A is 6 times the volume of solid B.

Given that $k > \sqrt[3]{7}$

find an expression for h in terms of k and r

$h = \dots\dots\dots$

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION**Q#: 24****Marks: 6****TOPIC: Area & Volume of Similar Shapes**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Small cone volume**

Small cone volume:

$$V_s = \frac{1}{3}\pi r^2 h$$

2. The larger similar cone hasThe larger similar cone has radius kr and height kh , so its volume is

$$V_L = \frac{1}{3}\pi(kr)^2(kh) = \frac{1}{3}\pi k^3 r^2 h$$

3. Calculate volume

So the frustum volume is

$$V_L - V_s = \frac{1}{3}\pi r^2 h(k^3 - 1)$$

4. The volume of one hemisphereThe volume of one hemisphere of radius r is

$$\frac{2}{3}\pi r^3$$

5. Solid A is the frustum

Solid A is the frustum plus one hemisphere:

$$V_A = \frac{1}{3}\pi r^2 h(k^3 - 1) + \frac{2}{3}\pi r^3$$

6. Solid B is the small

Solid B is the small cone plus one hemisphere:

$$V_B = \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$$

7. GivenGiven $V_A = 6V_B$:

$$\frac{1}{3}\pi r^2 h(k^3 - 1) + \frac{2}{3}\pi r^3 = 6\left(\frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3\right)$$

8. Divide byDivide by $\frac{1}{3}\pi r^2$:

$$h(k^3 - 1) + 2r = 6h + 12r$$

$$h(k^3 - 7) = 10r$$

$$h = \frac{10r}{k^3 - 7}$$

FINAL ANSWER

$$h = \frac{10r}{k^3 - 7}$$

EA

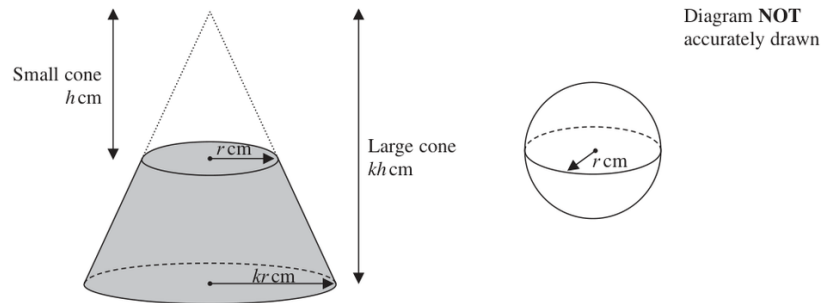
PAST PAPER QUESTION**Q#: 24****Marks: 6****TOPIC: Area & Volume of Similar Shapes**

Jun 2022 P2HR - Jun 2022 | P2HR

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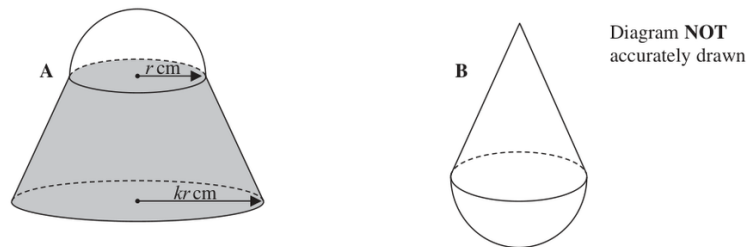
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Jun 2022 P2HR - Jun 2022 | P2HR

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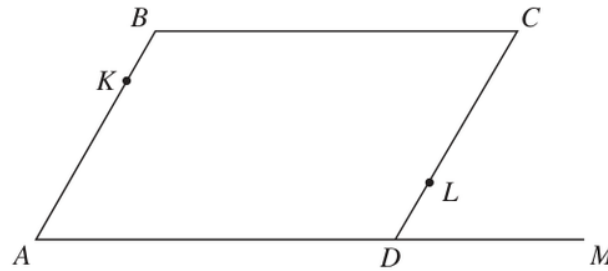
FINAL ANSWER

$$h = \frac{10r}{k^3 - 7}$$

EA

PAST PAPER QUESTION**Q#: 25****Marks: 5****TOPIC: Vectors**

Jun 2022 P2HR - Jun 2022 | P2HR

25 $ABCD$ is a parallelogram and ADM is a straight line.Diagram **NOT**
accurately drawn

$$\vec{AB} = \mathbf{a} \quad \vec{BC} = \mathbf{b} \quad \vec{DM} = \frac{1}{2} \mathbf{b}$$

 K is the point on AB such that $AK:AB = \lambda:1$ L is the point on CD such that $CL:CD = \mu:1$ KLM is a straight line.Given that $\lambda:\mu = 1:2$ use a vector method to find the value of λ and the value of μ

$$\lambda = \dots\dots\dots$$

$$\mu = \dots\dots\dots$$

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION**Q#: 25****Marks: 5****TOPIC: Vectors**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let be the origin**Let A be the origin.

$$\vec{AB} = \mathbf{a}, \quad \vec{BC} = \mathbf{b}$$

2. Read the graph

So

$$B = \mathbf{a}, \quad D = \mathbf{b}, \quad C = \mathbf{a} + \mathbf{b}$$

3. Read the graph

Also

$$D\vec{M} = \frac{1}{2}\mathbf{b}$$

4. Read the graph

So

$$M = \frac{3}{2}\mathbf{b}$$

5. Point is onPoint K is on AB :

$$K = \lambda \mathbf{a}$$

6. Read the graphPoint L is on CD , and $C\vec{D} = -\mathbf{a}$:

$$L = C + \mu(D - C)$$

$$L = \mathbf{a} + \mathbf{b} + \mu(-\mathbf{a})$$

$$L = (1 - \mu)\mathbf{a} + \mathbf{b}$$

7. Given

Given

$$\lambda : \mu = 1 : 2$$

$$\mu = 2\lambda$$

8. Read the graph

Since K, L, M are collinear, $\vec{KM}$ is a multiple of $\vec{KL}$.

$$\vec{KM} = -\lambda \mathbf{a} + \frac{3}{2} \mathbf{b}$$

$$\vec{KL} = (1 - \mu - \lambda) \mathbf{a} + \mathbf{b}$$

$$\vec{KL} = (1 - 3\lambda) \mathbf{a} + \mathbf{b}$$

9. Compare coefficients using

Compare coefficients using

$$\vec{KM} = \frac{3}{2} \vec{KL}$$

$$-\lambda = \frac{3}{2}(1 - 3\lambda)$$

$$-2\lambda = 3 - 9\lambda$$

$$7\lambda = 3$$

$$\lambda = \frac{3}{7}$$

10. Read the graph

Then

$$\mu = 2\lambda = \frac{6}{7}$$

FINAL ANSWER

$$\lambda = \frac{3}{7}, \quad \mu = \frac{6}{7}$$

PAST PAPER QUESTION

Q#: 25 Marks: 5

TOPIC: **Vectors**

Jun 2022 P2HR - Jun 2022 | P2HR

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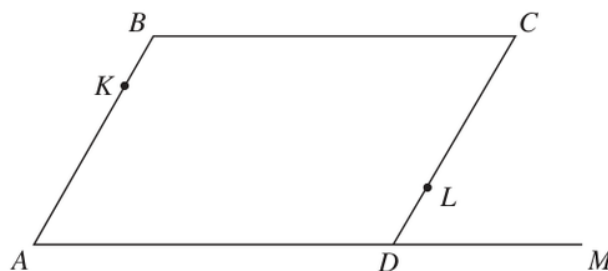


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$\lambda = \dots\dots\dots$

$\mu = \dots\dots\dots$

(Total for Question 25 is 5 marks)

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Jun 2022 P2HR - Jun 2022 | P2HR

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Then

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FINAL ANSWER

$$\lambda = \frac{3}{7}, \quad \mu = \frac{6}{7}$$